

Quick Check

Lesson 3: Counting Life: Characteristics of Populations and Direct Counting in the School Compound

10 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Ecology | Lesson 3

Villagers beside a Kenyan river say, "There used to be many fish." Which action would best turn this statement into evidence?

- A** Look at the river and guess how many fish are there
- B** Count the fish in a known area at different times and compare the numbers
- C** Ask more villagers whether they agree
- D** Wait to see if the smell of the water goes away

What do we call all the fish of one species living together in the same river?

A A community

B A habitat

C An ecosystem

D A population

A group marks a 10 m x 10 m area on the school field and counts 250 grass tufts. What is the population density of grass tufts?

A 25 per m²

B 2.5 per m²

C 0.4 per m²

D 350 per m²

During the jigsaw counting, why must the group agree on what counts as ONE plant before they start counting?

- A** So that the density will always be a whole number
- B** So that everyone counts in the same way and no plant is counted twice or missed
- C** So that the counting finishes faster
- D** So that the area marked will be larger

Site A has 300 weeds in a 10 m x 10 m area. Site B has 300 weeds in a 20 m x 10 m area. Which statement is correct?

- A** Both sites have the same density because they have the same number of weeds
- B** Site B has the higher density because its area is larger
- C** Site A has the higher density because the same number of weeds is found in a smaller area
- D** Density cannot be compared unless the number of weeds is different

Which situation is suitable for the direct counting method?

- A** Counting the trees growing in a small school plot
- B** Counting the insects flying over a whole forest
- C** Counting all the fish in Lake Victoria
- D** Counting all the algae cells in a river

Why would direct counting of every fish in a river be difficult?

- A** Rivers have no area that can be measured
- B** Fish do not belong to any population
- C** Fish density cannot be calculated using area
- D** Fish can move and hide in deep water, so they cannot all be seen and counted

A group counted grass in two 10 m x 10 m areas of the school field. A learner asks, "Do these two counts tell us how the grass is spread over the whole field?" What is the best answer?

- A** Not necessarily, because two small areas may not represent the whole field, so we need to test if the distribution is uniform
- B** No, because density is not affected by where the area is
- C** Yes, because any two areas will always show the whole field
- D** Yes, because direct counting always works for the whole field

A site has 180 plants in a 100 m² area. What is the population density?

A 0.18 per m²

B 18 per m²

C 1.8 per m²

D 80 per m²

Which statement correctly defines population density?

- A** The total number of species in an ecosystem
- B** The number of individuals counted in the whole habitat
- C** The number of individuals of a species per unit area
- D** The area occupied by one individual of a species only

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).